# Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 10 · SLO 1 — Solved Worksheet

**Name:** DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

## Unit â€“ 2 : Session â€“ 10 : SLO â€“ 1 | Doubly Linked List â€” Insertion at the Beginning and End

### I. Insert a New Node VAL at the Beginning

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 9
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET NEW_NODE -> PREV = NULL
- Step 6: SET NEW_NODE -> NEXT = START
- Step 7: SET START -> PREV = NEW_NODE
- Step 8: SET START = NEW_NODE
- Step 9: EXIT

### II. Insert a New Node VAL at the End

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 11
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET NEW_NODE -> NEXT = NULL
- Step 6: SET PTR = START
- Step 7: Repeat Step 8 while PTR -> NEXT != NULL
- Step 8: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 9: SET PTR -> NEXT = NEW_NODE
- Step 10: SET NEW_NODE -> PREV = PTR
- Step 11: EXIT